Who Gets Their Money Back?

Predicting outcomes and mapping themes in CFPB consumer complaints

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1. Introduction

A financial complaint can end with an explanation, a correction, or money back. Those are different outcomes. This project asks whether the complaint's story and basic details help predict the company's recorded monetary relief response, and what recurring themes appear beyond the official product and issue labels.

I chose this problem because the narratives explain what went wrong in a way that a product label cannot. Reading millions of them by hand is unrealistic. A useful model could help an analyst find patterns and choose complaints for closer review. It should not decide whether a person deserves help.

Part A compares logistic regression, boosted trees, and a small neural network using structured fields, TF-IDF text, and sentence embeddings. Part B compares NMF, LDA, k-means, and UMAP with HDBSCAN. The final experiment feeds topic weights back into outcome prediction. The contribution is the comparison across representations, the time-based evaluation, and the connection between readable themes and prediction.

The original strongest model was logistic regression with metadata and TF-IDF: test average precision was 0.332, against a 0.014 prevalence baseline. The additional validation search gives 0.334 with C = 0.05. NMF gives more coherent topics than LDA in the original topic-count search. Adding 15 topic weights improves the structured LightGBM's test average precision from 0.280 to 0.306. These are results on filed complaints with published narratives, not estimates for all consumers.

2. Related work

Bastani, Namavari, and Shaffer (2019) use LDA to find topics in CFPB narratives and follow their trends over time [1]. My project adds a direct comparison with NMF and embedding clusters, then checks whether topic features help predict monetary relief.

Vaishnav et al. (2024) combine complaint response prediction with LDA-based complaint analysis [2]. Their platform considers response timeliness and response categories. I focus on the narrower monetary relief label, compare several text representations, and keep recent years separate for evaluation.

Wang, Zhu, and Chen (2026) combine LDA topics, engineered text features, and categorical information in an XGBoost monetary relief model with a temporal split [3]. This is the closest match. My additional questions concern representation choice, historical validation, and topic stability. The scores across these studies are not directly comparable because the samples and evaluation setups differ.

This work began with the proposal and notebooks for this SIADS 696 project. The proposal states that it was not carried over from another course or earlier Milestone project. The cited studies informed the questions; their project code was not imported into the supplied notebooks.

3. Data sources and scope

The source is the CFPB Consumer Complaint Database [4]. The current bulk CSV is useful for broad exploratory checks, but the modeling data comes from the CFPB's fixed narratives archive [5]. The archive contains 21 ZIP files with CSV records, covering complaints received through August 14, 2026. The project stores the narrative rows from each export in Parquet files. The complete source URLs are recorded in the repository's data/sources.json.

The local bulk CSV is 5.47 GB. Its cached 0.5% exploratory sample has 90,312 rows; that sample is not the modeling dataset. The 21 narrative caches contain 3,851,415 records. For example, the caches contain 487,410 narratives from 2023, 814,385 from 2024, 1,222,050 from 2025, and 121,394 from the partial 2026 period. The 2026 count reflects incomplete coverage and should not be read as a full-year trend.

The important fields are Complaint ID, Date received, Product, Sub-product, Issue, Sub-issue, Company, State, Tags, Consumer complaint narrative, and Company response to consumer. The field reference explains these definitions [6]. The response is the target. Complaint ID identifies a record, and the received date defines time windows. Neither is a predictor.

Only response labels beginning with "Closed" are included. The target is 1 for "Closed with monetary relief" and 0 for every other included closure. "In progress" and "Untimely response" are excluded. Older generic "Closed" and "Closed with relief" responses do not tell us whether money changed hands; they remain in the historical topic sample as nonpositive labels. That limits comparisons of relief rates across early years.

Part A samples 20% of eligible 2018 to 2022 records, 10% of 2023 records, and 5% of 2024 to 2025 records, using seed 0. Sampling is proportional within each window. After cleaning, no empty text remains and Complaint IDs are unique.

Table 1. Supervised sample and time windows.

Window	Years	Complaints	Monetary relief	Relief rate
Training	2018 to 2022	191,186	6,983	3.65%
Validation	2023	48,688	1,203	2.47%
Test	2024 to 2025	101,568	1,440	1.42%

Part B takes up to 1,000 records from each product-group and year cell from 2015 to 2025. Removing 2,718 repeated normalized narratives and 5 empty normalized narratives leaves 97,585 documents. Its earliest received date is March 19, 2015. This balanced sample gives smaller product groups a chance to form themes instead of letting credit reporting dominate everything.

Each Part B row has a weight equal to its product/year population divided by the retained sample count in that cell. The weights restore those cell totals. They do not make deduplication random or turn these complaints into a representative sample of consumers. A repeated template may describe real, separate experiences.

4. Feature engineering and supervised methods

The shared cleaning function removes XXXX redaction masks, escaped export line breaks, leading byte-string markers, and stray export markers. It replaces redacted dollar expressions with the token dollaramount and collapses whitespace. This preserves a mention of money without trying to reconstruct a redacted amount. Lowercasing happens inside the text vectorizers.

Missing categorical values become the literal category None. In Part A, Tags is missing in 303,986 rows, Sub-issue in 27,968, State in 1,201, and Sub-product in 7. Missingness therefore remains visible rather than silently dropping those complaints. Product names from different taxonomy periods are mapped to ten stable product groups. Issue and sub-issue labels are not harmonized across time, so their changing vocabulary is still a limitation.

Table 2. Final feature representations. The appendix explains the full inventory embedded in this PDF.

Representation	Final features	Handling
Structured	Product group, sub-product, issue, sub-issue, company, state, tags	803 one-hot columns; categories seen fewer than 20 times pooled
TF-IDF	Word unigrams and bigrams	20,000 training-selected terms, min_df 10, max_df 0.9, sublinear term frequency
Combined	All structured columns plus TF-IDF	20,803 columns
Embeddings	MiniLM coordinates 0 through 383	384 normalized values; first 256 word pieces of each narrative
Topic features	NMF weights 0 through 14	15 nonnegative values in the integration experiment

The supervised vectorizers and category encoders are fitted on training data only. Validation and test data only pass through transform. Rare or unseen categories go into the infrequent group when that group exists; otherwise unseen categories produce all-zero columns for that field. Post-response fields, including Company public response and Timely response?, are excluded to avoid using the answer as an input.

Logistic regression learns an additive score with L2 regularization. It is a sensible baseline for many sparse text features and makes individual terms easy to inspect. The original setting is C = 0.25 with balanced class weights. The added sensitivity search compares C values 0.01, 0.05, 0.25, 1, and 4 using 2023 average precision.

LightGBM combines decision trees and can capture interactions among company, issue, and text features [8]. The original grid uses 500 trees, learning rate 0.05, 63 leaves, row sampling 0.8, feature sampling 0.3, and balanced class weights. It is tested on each representation. These settings are fixed in the historical folds rather than selected separately on each fold's validation records.

The neural model is an MLP with one 256-unit hidden layer on the MiniLM embeddings. Its nonlinear interactions test whether the compressed text representation gains from a neural classifier. It uses early stopping and seed 0. It does not use the class weighting applied to the other two families. That makes this a comparison of practical pipelines, not a controlled comparison of model mechanisms alone.

The embeddings come from the public all-MiniLM-L6-v2 model [7]. Long stories are truncated, which may lose the part that explains the dispute. The original grid uses the MLP only on embeddings; it does not train an MLP on every sparse representation. A constant prior-score classifier provides a ranking baseline. Scikit-learn supplies the common modeling and evaluation tools [9].

5. Supervised evaluation

Average precision, abbreviated AP, is the primary metric. The notebooks call it PR-AUC; the actual function is scikit-learn's average_precision_score, a recall-weighted summary rather than trapezoidal integration [10]. It puts attention on finding a rare positive class. A constant-score baseline has AP equal to the positive rate, which is 0.014 on the test sample. Accuracy would reward predicting no monetary relief for almost everyone.

ROC-AUC describes ranking across both classes, but can look strong while precision is still modest. F1 summarizes one threshold's balance of precision and recall. The threshold is selected using 2023 labels only. Precision among the top 1% simulates a fixed review budget. The constant model cannot meaningfully order ties, so its top-1% value is not an operational baseline.

Table 3. Original representation grid. These are validation and held-out test results, separate from the repeated validation in Table 4.

Features	Model	Val AP	Test AP	Test F1	Top 1% precision
structured	Majority	0.025	0.014	0.028	Tie-dependent
structured	LogReg	0.295	0.270	0.351	0.359
structured	LightGBM	0.309	0.280	0.368	0.323
tfidf	LogReg	0.326	0.290	0.340	0.367
tfidf	LightGBM	0.298	0.264	0.346	0.345
embedding	LogReg	0.268	0.243	0.338	0.312
embedding	LightGBM	0.300	0.270	0.353	0.332
embedding	MLP	0.316	0.277	0.344	0.357
structured+tfidf	LogReg	0.365	0.332	0.383	0.407
structured+tfidf	LightGBM	0.324	0.288	0.362	0.384

The best original family representatives by 2023 AP are combined-feature logistic regression, combined-feature LightGBM, and the embedding MLP. To avoid relying only on one split, the added evaluation uses five expanding time folds within the 2018 to 2022 training window. It splits complete received dates, keeps training dates earlier than validation dates, and rebuilds the learned features inside each fold.

Table 4. Historical cross-validation. SD is the sample standard deviation over five folds; model settings are the original fixed configurations.

Model	Features	CV AP (mean +/- SD)	CV ROC-AUC (mean +/- SD)	Mean fit (s)
LightGBM	structured+tfidf	0.321 +/- 0.016	0.926 +/- 0.019	183.8
LogReg	structured+tfidf	0.367 +/- 0.021	0.940 +/- 0.017	3.4
MLP	embedding	0.328 +/- 0.022	0.926 +/- 0.017	13.4

These folds have different class rates and training sizes, so their SD describes variation across time windows. It is not a confidence interval. The 2023 representation selection is separate from these earlier folds. The C search below is a later validation exercise; Table 4 continues to describe the original C = 0.25 logistic model. The final 2024 to 2025 test window is kept out of fitting and threshold selection.

Metadata and text work better together: adding metadata to TF-IDF raises original test AP from 0.290 to 0.332. Logistic regression also fits much faster than LightGBM across all five folds; timings exclude feature preparation. The compact embeddings do not win here. That comparison depends on the classifier, truncation limit, and task.

6. What the best model uses

The added C search selects C = 0.05. The following importance and removal checks use that combined-feature logistic model, rather than borrowing an explanation from a different model. Its test AP is 0.334. A term's signed coefficient shows how it changes the model score while other encoded inputs stay fixed. It is not evidence that using that word changes a company's decision.

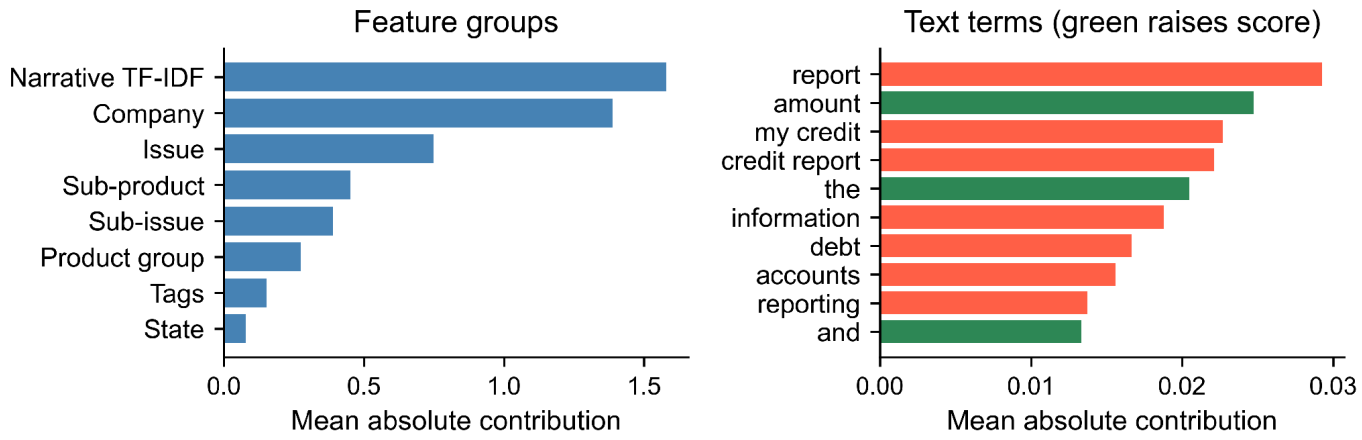


Figure 1. Mean absolute feature contribution in the selected combined logistic model. Contributions are coefficient times observed feature value, measured on test records and grouped by field. They describe this fitted model, not independent causal effects. Green text bars raise the score; red bars lower it.

The comparison also needs removal and retraining. A large coefficient alone can be misleading when several inputs carry the same information. For example, company names appear both in the Company field and in narratives. Product, sub-product, and issue also overlap. Removing one group allows the remaining features to compensate.

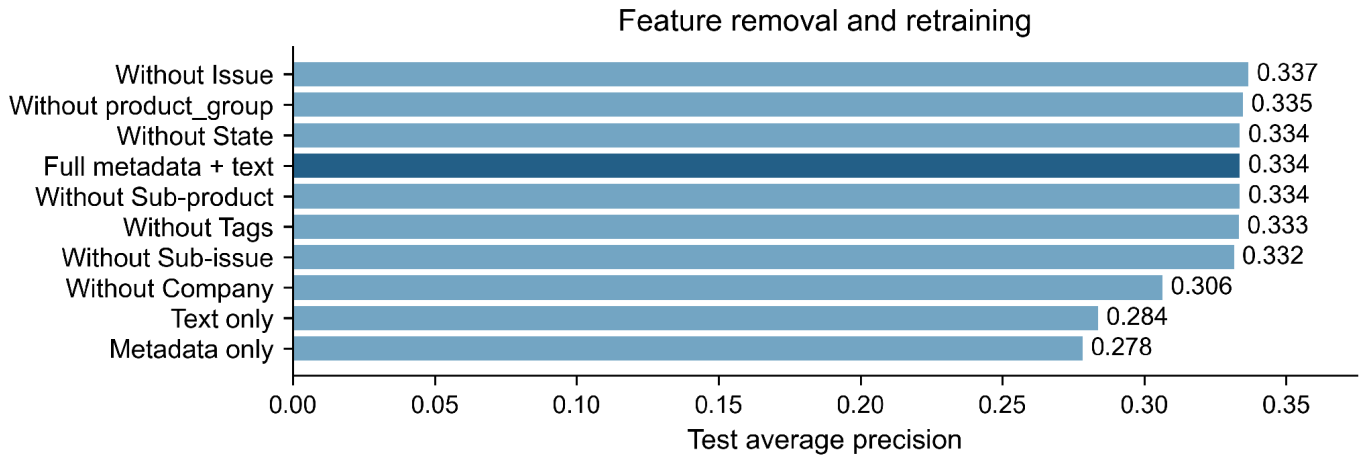


Figure 2. Feature ablation with the selected C held fixed. Each reduced model is retrained on 2018 to 2022 data. The full model has test AP 0.334, metadata alone 0.278, text alone 0.284, and the full model without Company 0.306.

Removing Company lowers AP from 0.334 to 0.306, the largest loss from removing a single structured field. Removing Issue slightly improves test AP to 0.337, while its validation AP is lower. I keep the full model selected before that test comparison. This points to overlap among fields rather than a claim that every field helps.

Importance describes the fitted score; ablation measures what retraining loses. Other fields may replace a removed input. The earlier notebook's structured-tree SHAP figure is a secondary view, not the explanation of the winning logistic model.

7. Sensitivity, calibration, and review tradeoffs

The regularization search uses the full 2018 to 2022 training window and evaluates five C values on 2023. Validation AP ranges from 0.320 to 0.367, with C = 0.05 selected. Only after that selection is the chosen model evaluated on the test years. This checks sensitivity to regularization without selecting C using the test score.

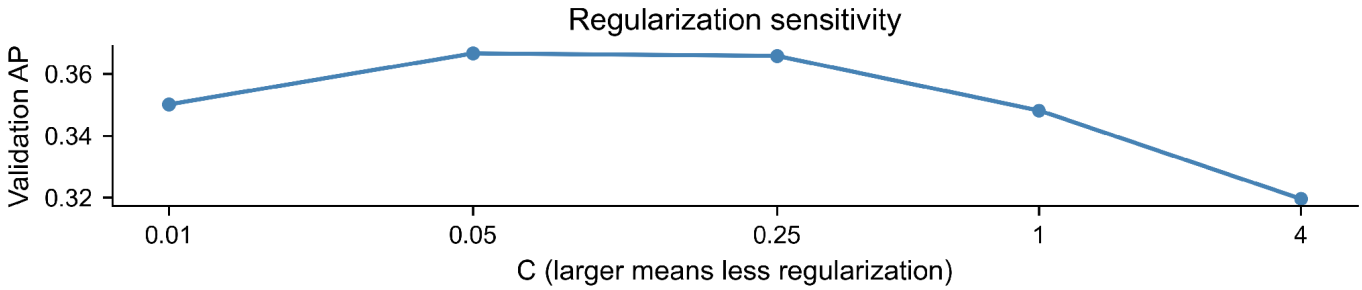


Figure 3. Validation AP across the regularization settings. C controls the strength of the L2 penalty; smaller values shrink the coefficients more.

Balanced class weighting helps the classifier find rare relief cases, but its raw scores are not calibrated probabilities. The calibration check fits a logistic mapping of raw log-odds to outcomes using 2023 only, then applies that mapping to 2024 and 2025. The yearly results and Brier scores are included in the appendix.

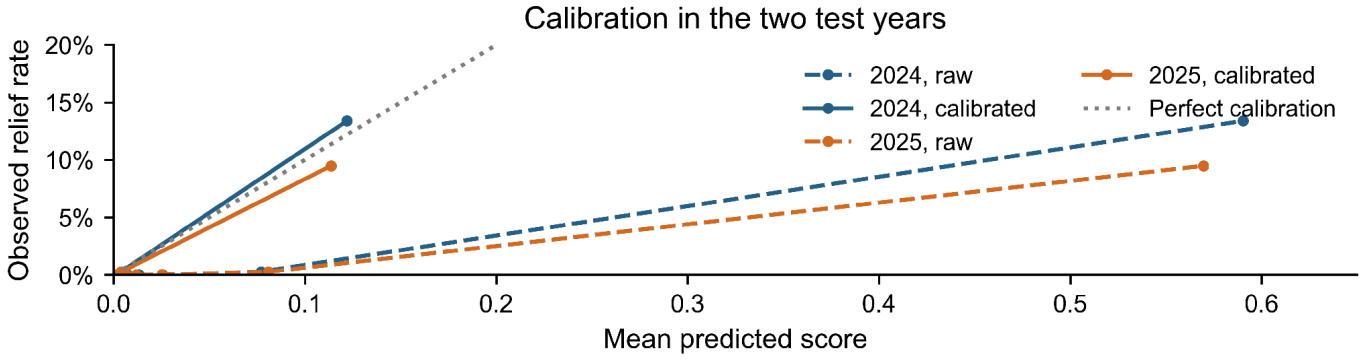


Figure 4. Raw weighted scores and scores calibrated on 2023, evaluated separately in 2024 and 2025. A point on the diagonal would match the observed rate in that score bin. Calibration can still drift when the complaint mix and relief rate change.

Table 5. Test review policies for the selected logistic model. Top-fraction policies use a fixed capacity, not test labels, to choose records.

Review policy	Reviewed	Precision	Recall	F1
Validation max-F1	3333	0.271	0.628	0.379
Raw score 0.50	7378	0.176	0.901	0.294
Review top 0.5%	507	0.475	0.167	0.248
Review top 1.0%	1015	0.403	0.284	0.333
Review top 2.0%	2031	0.320	0.451	0.375
Review top 5.0%	5078	0.225	0.794	0.351

At the validation F1 threshold of 0.824, precision is 27.1% and recall is 62.8%. The true negative, false positive, false negative, and true positive counts are 97,699 / 2,429 / 536 / 904. Smaller queues trade recall for precision. Set that budget for the analyst's needs and keep human review available.

8. Failure analysis and change over time

Example 1: Earlier refund confused with the final outcome. Complaint 12230857, received 2025-02-26, Checking/savings. The recorded outcome was Closed with explanation. Its score was 0.990, so it was a false positive at the validation threshold. The narrative describes one overdraft fee and a maintenance fee that had already been refunded, followed by a request to waive another fee. Words about refunds and fees resemble successful cases, even though an earlier refund is different from the final response to this complaint. An event-order representation could separate prior refunds from the unresolved request.

Example 2: Too little information in the narrative. Complaint 16791143, received 2025-10-23, Money transfer. The recorded outcome was Closed with monetary relief. Its score was 0.023, so it was a false negative at the validation threshold. The cleaned text is only a short fragment naming Chime and saying it authorized something. There is almost no detail about the transaction, dispute, or requested remedy. The company prior and the few available words cannot explain this positive outcome. A model could flag low-information text for human review, rather than giving a confident recommendation.

Example 3: Product label hides a different financial issue. Complaint 11405369, received 2025-01-06, Credit reporting. The recorded outcome was Closed with monetary relief. Its score was 0.002, so it was a false negative at the validation threshold. The record is labeled credit reporting, but the story also describes a compromised Chime account and missing funds after an alleged data breach. The credit-reporting category usually has little monetary relief. A linear model may underweight the bank-account loss described inside a much broader identity-theft narrative. A multi-issue text representation could distinguish credit-report correction from a request to recover missing funds.

The examples above are actual records from the saved test sample, selected to illustrate different possible failure mechanisms. The explanations are interpretations of the available text. The public data does not reveal the internal investigation, evidence, or payment decision, so the reason for an individual error cannot be proven from a narrative alone.

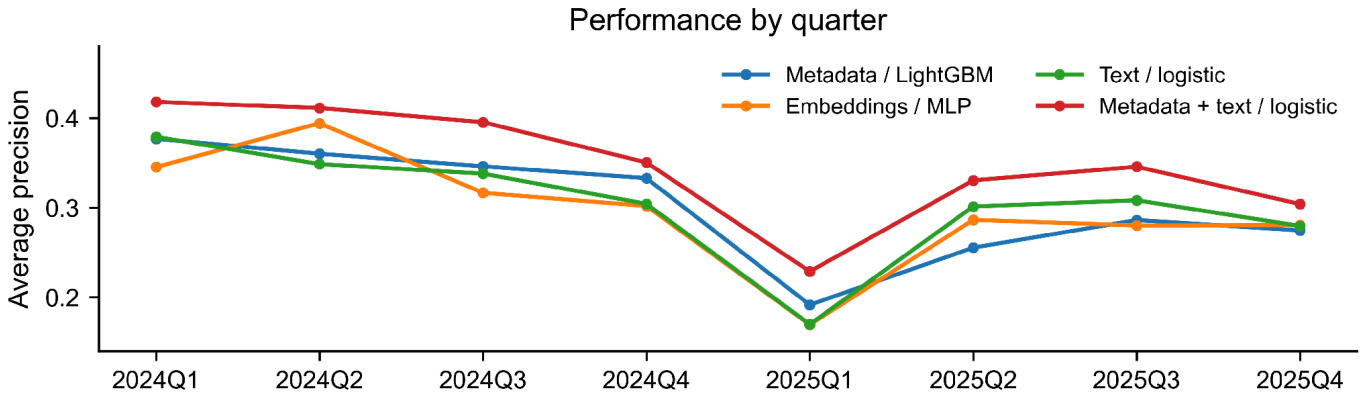


Figure 5. Original test AP by quarter. The combined model falls from 0.418 in 2024 Q1 to 0.229 in 2025 Q1, then partly recovers. Every quarter remains a separate slice of the same held-out window.

The 2025 Q1 money-transfer sample contains 2,465 complaints, versus 122 in 2024 Q1. Its relief rate is 0.6%, and 60.3% name Block, Inc. The corresponding 2024 Q1 figures are 13.1% and 13.9%. That shift is consistent with a change in complaint mix contributing to the dip; it does not prove that one company or a template caused it. AP also changes with prevalence.

Exact normalized text matching finds 5023 test narratives that also appear in training. Removing those matches only for a diagnostic gives test AP 0.334. The official test set stays unchanged. Near-duplicate text and repeated consumers may remain because there is no consumer identifier and this audit only checks exact normalized matches.

9. Unsupervised methods and evaluation

Part B uses the same cleaned text but a different sample and vocabulary. Count features use English stop words plus 15 selected company-name words, alphabetic tokens of at least three letters, unigrams and bigrams, min_df 20, max_df 0.5, and a 20,000-term limit. Removing a few brands helps avoid topics that are just a company name. It does not remove every brand or make company comparisons independent of text.

NMF factors nonnegative TF-IDF into document weights and topic terms. It favors additive, readable groups of words. **LDA** is a probabilistic mixture of topics over word counts; it uses counts rather than TF-IDF. Both compare 10, 15, 20, 25, and 30 topics. The original search selects 15 for each based on top-ten-term NPMI coherence. The added NMF run checks topic-count sensitivity again and also varies initialization.

k-means groups the normalized 384-dimensional MiniLM embeddings around 15 centers, with three initializations. It assigns every document. **UMAP plus HDBSCAN** first reduces the embeddings to five dimensions, using 15 neighbors, cosine distance, and min_dist 0. It then finds dense groups with minimum cluster size 500 and min_samples 10. It can leave documents unassigned. These embedding methods provide a different view from word-count topic models [11, 12].

Table 6. Original unsupervised comparison. Silhouettes use cosine distances in the original embedding space. NMI measures alignment with existing labels; higher alignment is not automatically a better discovery.

Method	Groups	Noise	NMI product	NMI issue	Cosine silhouette	Bootstrap ARI
NMF	15	0.0%	0.340	0.272	0.022	0.717
LDA	15	0.0%	0.313	0.270	0.008	Not run
k-means	15	0.0%	0.375	0.302	0.063	0.730
UMAP+HDBSCAN	32	43.4%	0.476	0.406	0.090	Not run

NPMI measures how often a topic's leading terms appear together in documents. It helps assess readability, but overlapping unigrams and bigrams can raise it mechanically. Silhouette measures separation in embedding space and therefore favors methods built in that space. Bootstrap ARI measures agreement between refitted labelings on the same documents. The original three bootstrap refits were run for NMF and k-means only. Missing ARI values do not mean zero stability.

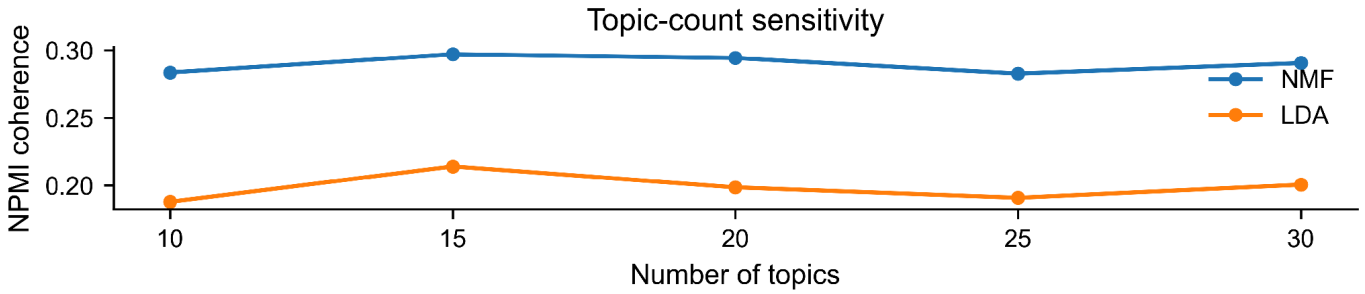


Figure 6. Original NPMI sensitivity to topic count. NMF scores exceed LDA scores at every tested count. NMF peaks at 15 topics with 0.297; LDA also peaks at 15 with 0.214. These values rank this experiment's outputs, not the overall usefulness of the algorithms.

10. NMF: readable themes across the complaint mix

NMF is the preferred model for the main theme analysis because it produces readable terms and covers every document. It does not have the best embedding silhouette. That is acceptable here: the goal is a useful topic summary, not only compact embedding clusters. The original bootstrap ARI of 0.717 suggests meaningful agreement with room for variation.

The 15 themes include debt collection, vehicle loans, student loans, mortgage servicing, card charges, bank accounts, identity theft, and credit-report disputes. Some themes capture writing style or repeated wording rather than a single financial problem. For instance, the customer-contact theme is broad, while the legal-section and Cash App themes are more tied to a particular style of complaint. Those themes need to be read before they are given stronger labels.

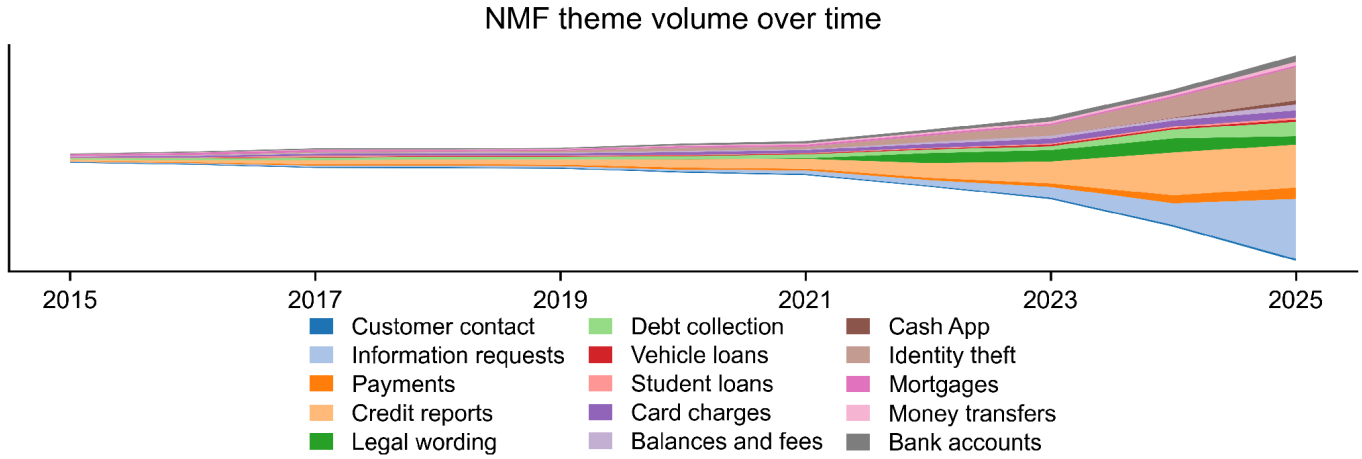


Figure 7. NMF theme volume in the added run, weighted to product/year totals in the narrative archive. The streamgraph shows volume, not the chance of relief or the share of all financial customers. Its vertical centering is only a plotting choice.

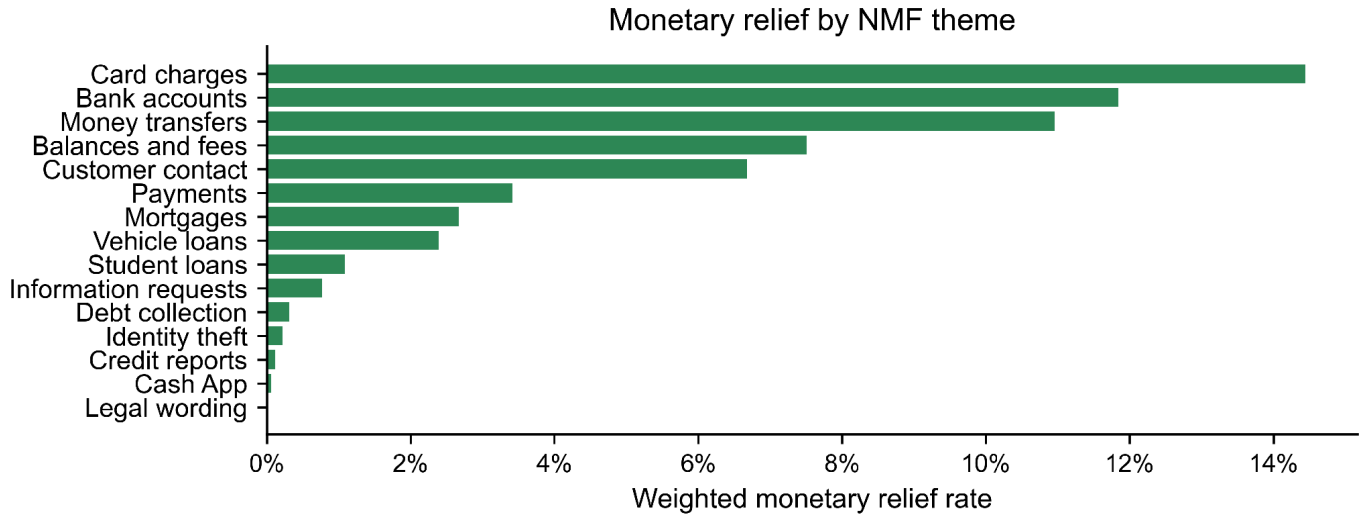


Figure 8. NMF theme relief rates in the added run, using the product/year weights. The appendix lists every theme and its sample count. The estimates pool years and reflect the retained, deduplicated sample.

The added search peaks at 20 topics, with NPMI 0.292 versus 0.291 at 15. That small gap supports keeping the simpler 15-topic summary, but it also shows that the original optimum is not exact. Random-initialized runs have ARI 0.508 to 0.567 against the fixed-initialization model, with NPMI 0.291 to 0.292. Labels and boundaries are useful summaries, not ground truth.

11. LDA: a probabilistic comparison

LDA models a document as a mixture of topics and a topic as a distribution over words. It can describe mixed narratives naturally, but the plots below simplify each document to its largest topic weight. That assignment discards information about secondary topics. The model uses online learning with batches of 4,096 documents and seed 0.

LDA's best original coherence is 0.214 at 15 topics, below NMF's 0.297. Its original product NMI is 0.313 and issue NMI is 0.270, close to NMF's alignment. Its original embedding silhouette is only 0.008. A low silhouette here means the assigned topics overlap in MiniLM space; it does not make the probabilistic fit invalid.

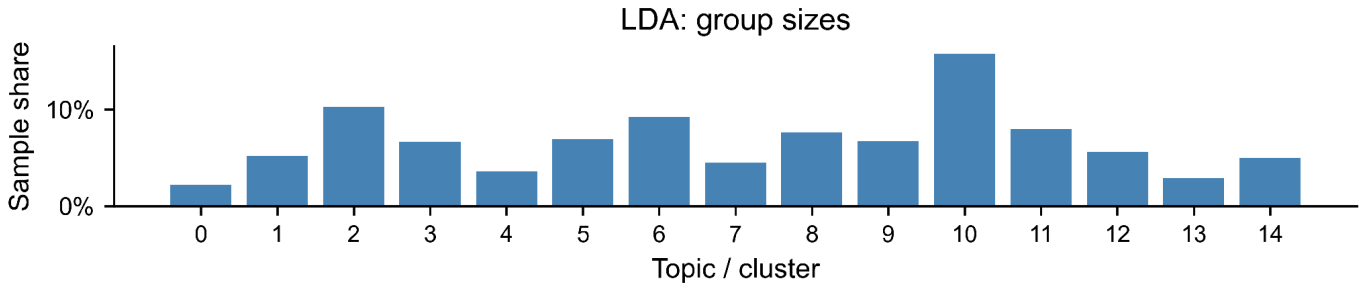


Figure 9. Largest-topic assignments in the added 15-topic LDA run. Fractions describe the balanced, deduplicated sample, without population weights. Topic numbers are internal labels.

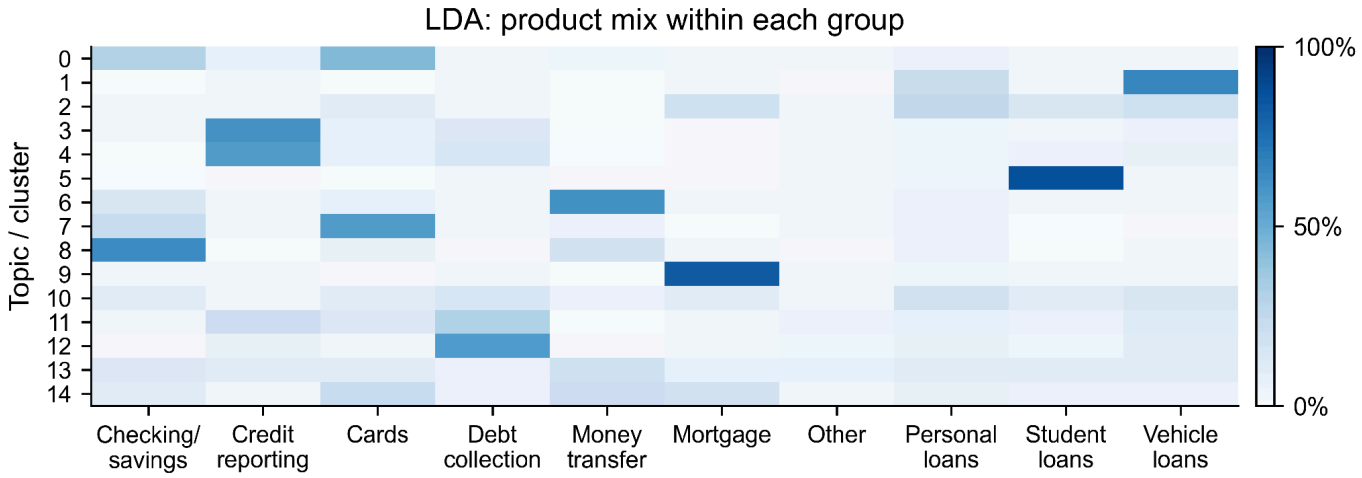


Figure 10. Product composition within each LDA topic in the added run. Each row sums to one. A topic can contain several products, which is useful when a complaint process crosses the official taxonomy.

These two views help check whether LDA finds reasonably sized groups and whether it mostly repeats product labels. The actual top-ten terms for every topic are saved in the results and the PDF attachment inventory includes the vocabulary. LDA is kept as a comparison rather than chosen for the main narrative because its top words are less coherent under the chosen measure. No formal blinded human-rating exercise was completed.

For a stronger follow-up, I would compare held-out likelihood and a small, independently rated set of representative complaints. That would test something different from word co-occurrence. More topic counts alone would not answer whether the themes are useful to a person reading complaints.

12. k-means: groups in sentence-embedding space

k-means uses distance to cluster centers rather than a word-generation or factorization model. With normalized embeddings, squared Euclidean distance is closely tied to cosine similarity. It is a straightforward way to test whether similar-sounding complaints form useful groups without building a topic model.

The number of clusters is 15 to make the comparison with the selected topic models easier. This is not a separately optimized k-means cluster count. Three initializations are used to reduce dependence on one starting configuration. All 97,585 documents receive a cluster, including documents that may sit between themes.

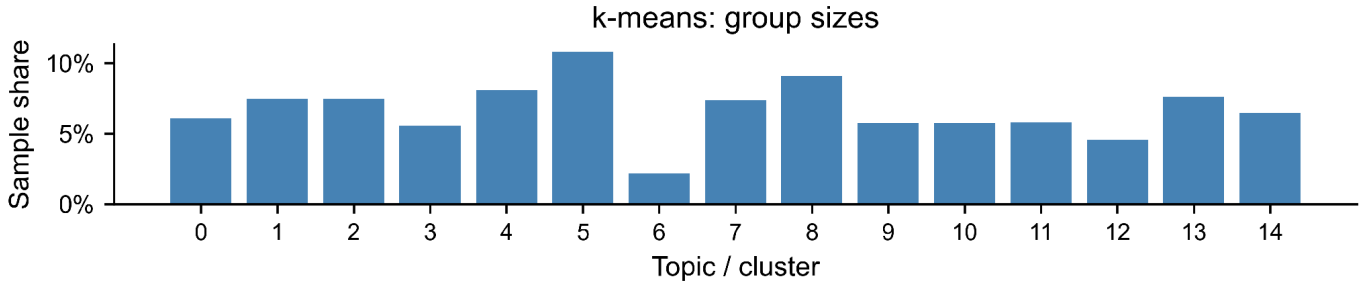


Figure 11. Cluster fractions from the added 15-cluster k-means run. These are sample fractions. Forced assignment gives full coverage but can hide ambiguous cases.

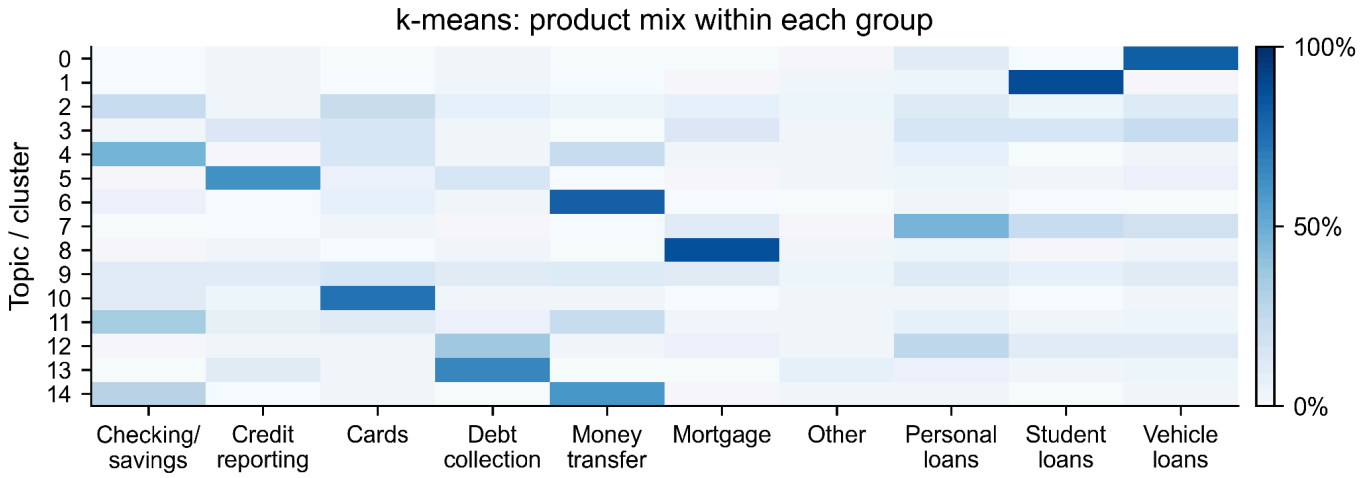


Figure 12. Product composition inside each k-means cluster. Rows sum to one. Product concentrations show which clusters mostly repeat a product distinction and which mix several products.

The original cosine silhouette is 0.063, higher than either classical topic model. Product NMI is 0.375 and issue NMI is 0.302. The original bootstrap ARI is 0.730, similar to NMF. These results suggest that the embedding clusters are a reasonable alternative for grouping complaints, with slightly stronger alignment to the official taxonomy.

The tradeoff is explanation. A center in 384 dimensions is harder to name than a short list of topic words. A future review should show several representative narratives per cluster and allow multiple labels. The current results support k-means as a grouping tool, but they do not establish that each center corresponds to one distinct consumer problem.

13. UMAP and HDBSCAN: denser groups, less coverage

HDBSCAN can discover clusters of different shapes and mark low-density records as noise. The five-dimensional UMAP representation makes that search practical. The separate two-dimensional plot below is only a view of 20,000 sampled records; it is not the space used to fit HDBSCAN.

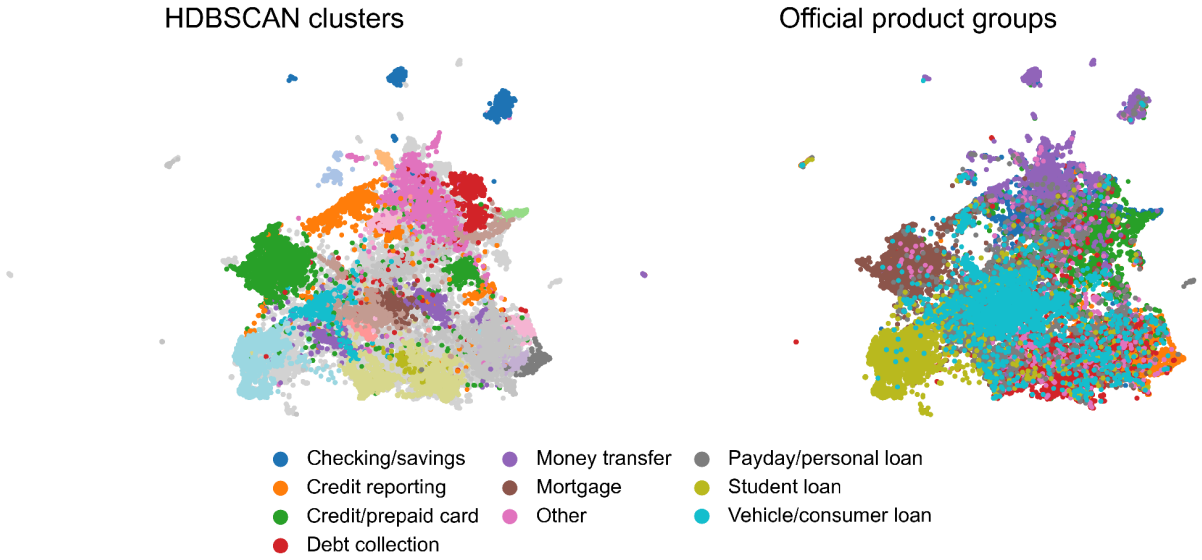


Figure 13. Two-dimensional UMAP view, colored by HDBSCAN labels from the added run on the left and product groups on the right. Gray means unassigned. Apparent distance and island size depend on the projection and should not be read as exact measures of financial similarity.

The original run finds 32 clusters and leaves 43.4% of documents as noise. Its product NMI of 0.476, issue NMI of 0.406, and cosine silhouette of 0.090 are the largest in the original comparison. However, those scores exclude the noise records. They therefore describe a selected 56.6% of the sample, while the other methods cover everyone.

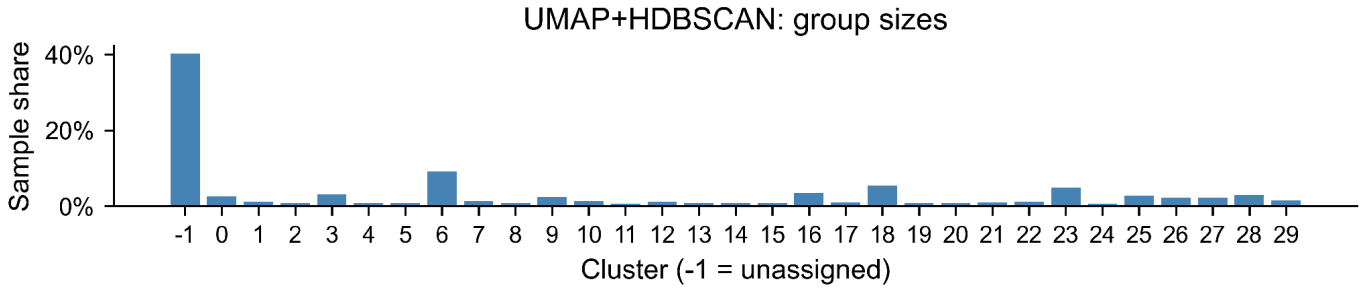


Figure 14. Cluster sizes in the added HDBSCAN run, including the unassigned group at -1. Exact labels and counts can vary with library versions and the UMAP fit. The saved assignment file distinguishes this added run from the original comparison.

This coverage tradeoff is why I did not pick HDBSCAN as the main summary of the complaint population. It may be useful for finding compact subgroups, but ignoring nearly half the narratives would make an incomplete theme map. Noise does not mean a complaint is unimportant, invalid, or unusual in a harmful way.

The added run finds 30 clusters with 40.2% noise, rather than exactly matching the original 32 and 43.4%. The broad coverage tradeoff survives, but exact cluster boundaries do not. The settings were not searched extensively, and no density-specific validity score was computed. A useful extension would vary neighborhood size and minimum cluster size while comparing stability and coverage together.

14. Connecting themes to prediction

The integration experiment uses a new topic pipeline fitted only to the Part A training text. It appends 15 NMF weights to the structured inputs and keeps the original LightGBM settings fixed. Validation and test narratives do not enter that fit. The topic count itself came from the broader Part B exploration, so this is an exploratory integration result rather than a completely isolated model-selection experiment.

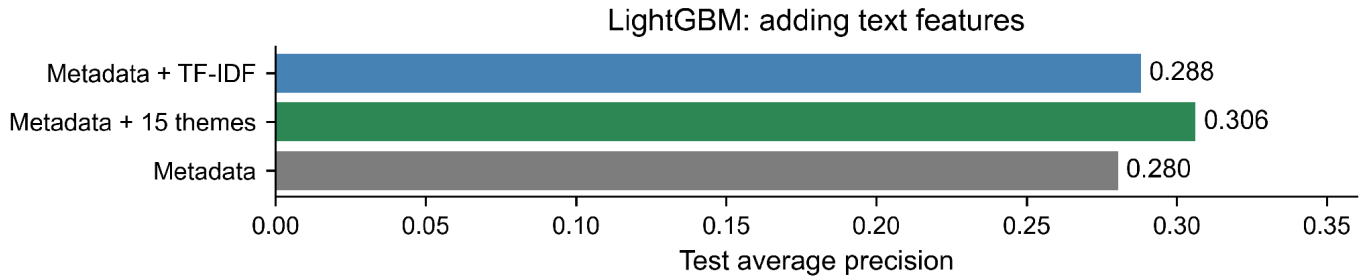


Figure 15. Original LightGBM integration results on the same test window. Structured features score 0.280 AP, structured features plus 15 topic weights score 0.306, and structured features plus 20,000 TF-IDF terms score 0.288. The topic model used for prediction is refitted on training data, so its topic numbers need not match the descriptive Part B topics.

The 0.026 AP gain is useful evidence that a small text summary can complement structured inputs for this tree model. It does not beat the best original combined logistic model at 0.332. Also, these integration variants were not compared over repeated folds, so the difference is a held-out observation, not a claim of statistical superiority.

The company/theme comparison in the original notebook also shows variation within a topic. Among sampled late-payment complaints, the recorded relief rates are about 33.3% at American Express, 31.5% at Citibank, 12.4% at Capital One, and 12.0% at JPMorgan Chase. These cells have at least 30 records. They are raw, unweighted comparisons within a stratified sample, with no adjustment for year, issue, or severity. They should not be used as a ranking of company fairness.

15. Discussion

What I learned from Part A. The useful surprise is that the simpler sparse linear model beats the more complex text pipelines. Metadata and text each contain information the other misses. The main challenge is that relief becomes rarer and the product mix shifts: credit reporting rises from about 57% of training records to 76% of test records. Stable product groups and time-based checks help expose that change. They do not make it disappear.

The next supervised steps would be better handling of repeated templates, calibration on more recent windows, a group-aware split if a safe consumer identifier were available, and uncertainty estimates that account for repeated complaints. The feature ablations also suggest a useful deployment question: how much performance is worth relying on company history, which may change or encode past disparities?

What I learned from Part B. Topic quality depends on the question. NMF is easier to describe, k-means forms cleaner embedding groups, and HDBSCAN improves separation while dropping many records. Another surprise is that repeated legal or campaign-style language becomes a theme of its own. That can help explain a volume shift, but it can also distract from the underlying dispute. The sample design, weights, and explicit coverage reporting help make those limits visible.

16. Limits and ethical considerations

The supervised target is a company-reported closure label. It does not measure the amount returned, whether the company was at fault, whether a consumer was satisfied, or whether non-monetary relief solved the problem. A model trained on that label can reproduce the company's historical response patterns without measuring what would be fair.

A practical use would be to support an analyst's review queue with human oversight. A low predicted score should not block a complaint, delay help, or advise a person to give up. Company, state, and tags can act as proxies for differences in access or treatment. The current work does not include a fairness audit across demographic groups; demographic data is limited and tags are mostly missing. Before use, I would audit error rates where the data supports it and keep a route for reviewing lower-scored cases.

The unsupervised outputs have their own risks. A cluster label can sound more authoritative than the evidence behind it. Repeated wording should not be called fraud, proof of automation, or a coordinated campaign without separate evidence. HDBSCAN noise should never be treated as a low-quality complaint. Company comparisons need adjustment for complaint mix and time before they can support stronger claims.

The CFPB warns that the database is not a statistical sample of consumers' experiences [4]. Only published narratives are modeled here, which adds an opt-in filter. Public redaction reduces privacy risk but does not remove it. The report uses short paraphrases for the failure cases, avoids extra personal details, and does not attempt to identify complainants.

The proposal included several possible extensions that were not completed: FDIC institution-size matching, lemmatization, MinHash near-duplicate removal, a formal blinded human-labeling study, BERTopic, and density-specific cluster validity. The final work uses normalized exact-text deduplication in Part B and an exact-text overlap audit in Part A. Those choices keep the scope manageable, but they leave meaningful work for a follow-up.

17. Statement of work and attribution

Garret Obenchain is the sole team member. He developed the supplied proposal, data preparation, exploratory analysis, supervised and unsupervised notebooks, initial evaluation, and plots. The final project also includes Codex assistance for added validation, feature ablation, sensitivity and calibration checks, failure-case analysis, code packaging, documentation, and report drafting. This statement identifies that assistance rather than presenting it as a separate human team member.

The data is from the CFPB. The analysis uses the cited open-source tools and the public MiniLM model. The three related studies informed the project framing; their code was not copied into this project. The earlier standup and proposal are stages of this same course project. No external collaborator is listed in the supplied materials.

18. Project materials

The repository contains the report PDF, analysis notebooks, results, plots, source URLs, and a 100-record data sample. The full data and embeddings are excluded because of size; the public archive links identify the inputs.

Repository: gobenchain/SIADS696-Final-Report. It is private, so the coach needs access.

The full feature inventory is in the repository and is also attached to the PDF. References and the appendix follow.

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Appendix A. Complete feature specification

The complete feature inventory lists every encoded supervised column and every unsupervised vocabulary term by index, with its source field. A copy is attached to the PDF. The same inventories are stored in results/feature_manifest.csv and results/b_feature_manifest.csv. Embedding coordinates are named embedding_000 through embedding_383; topic weights are nmf_000 through nmf_014. This keeps the full feature list inside the submitted PDF without adding hundreds of lines of vocabulary to the narrative.

The seven structured source fields are product_group, Sub-product, Issue, Sub-issue, Company, State, and Tags. A category is encoded as a 0/1 indicator, with training categories occurring fewer than 20 times combined into an infrequent category. Missing values become None. The complete source-name mapping from historic Product values to the ten product groups is in data_preparation.ipynb.

The supervised word tokenizer accepts alphabetic tokens of at least two letters. The vocabulary includes unigrams and bigrams appearing in at least ten training documents and no more than 90% of training documents, capped at 20,000 terms by frequency. Term frequency is 1 + log(count) for positive counts; inverse document frequency is smoothed; each row is L2 normalized. The combined model concatenates that representation with the one-hot fields.

The unsupervised vocabulary instead requires at least three letters per token, at least 20 sample documents, and no more than 50% of sample documents. It also removes English stop words and these brand tokens: wells, fargo, chase, jpmorgan, boa, america, citi, citibank, synchrony, navient, paypal, coinbase, equifax, experian, transunion. NMF uses sublinear TF-IDF of this count matrix; LDA uses the raw counts.

MiniLM embeddings contain 384 L2-normalized values. The model truncates text beyond 256 word pieces. k-means operates directly on those values. HDBSCAN uses UMAP dimensions 0 through 4, derived from those values. The two-dimensional UMAP display is a separate visualization and is not a classifier input.

Table 7. NMF topics in the added run. IDs are internal to this fit; they need not match a topic pipeline refitted on another sample.

Theme	Top terms	Sample n	Weighted relief
0	called, told, phone	5888	6.7%
1	information, request, provide	7157	0.8%
2	payment, late, payments	7789	3.4%
3	report, credit, credit report	8583	0.1%
4	section, section states, reporting	905	0.0%
5	debt, collection, collect	7285	0.3%
6	car, vehicle, loan	7005	2.4%
7	loans, loan, student	7652	1.1%
8	card, credit card, credit	9400	14.4%
9	amount, balance, paid	5748	7.5%
10	cash app, app, cash	604	0.1%
11	identity, theft, identity theft	3308	0.2%
12	mortgage, home, loan	8305	2.7%
13	money, transfer, transaction	7767	11.0%
14	account, bank, checking	10189	11.8%

Appendix B. Supplementary checks and reproducibility

Table 8. Calibration by test year. The Brier score measures squared probability error; lower is better. Its value is affected by the rare positive rate. A prevalence-only predictor can have a low score while providing no ranking value.

year	method	brier	mean_score	relief_rate
2024	Raw weighted score	0.0425	0.090	0.017
2024	2023 calibrated	0.0126	0.016	0.017
2025	Raw weighted score	0.0435	0.088	0.012
2025	2023 calibrated	0.0099	0.015	0.012

Calibration lowers the 2025 Brier score from 0.0435 to 0.0099. The mean calibrated score is still 1.50%, above the observed 1.22% relief rate. The mapping helps but does not fully track the later change in prevalence.

Table 9. Added NMF topic-count sensitivity. Reconstruction error is measured in the fitted TF-IDF space and normally falls as model capacity increases. It is not sufficient on its own to choose the topic count.

k	npmi	reconstruction_error	product_nmi
10	0.283	304.1	0.272
15	0.291	302.6	0.347
20	0.292	301.5	0.297
25	0.281	300.5	0.262
30	0.282	299.6	0.261

The original unsupervised summary uses a 10,000-document silhouette sample. The added run uses 3,000 documents to provide an affordable reproducibility check, recorded in b_reproduced_comparison.csv. These are separate measurements, not interchangeable estimates. The report keeps the original summary clearly identified and labels the newly produced charts as added runs.

The original and added runs both use seed 0 where stated. The NMF initialization check additionally uses seeds 0, 1, and 2 with random initialization. The original notebooks do not record a package-version manifest. The added topic and cluster fits do not exactly reproduce their original boundaries. Fixed seeds do not guarantee identical results across package versions and thread settings. The environment file and input checksums record the final run's context; they do not establish which difference caused the changed fits.

The supervised analysis and its plots are in part_a_supervised.ipynb. Topic modeling, clustering, and their plots are in part_b_unsupervised.ipynb. Shared loading and cleaning functions are in data_preparation.ipynb. The EDA notebook uses the current bulk CSV; that changing snapshot may differ from the exploratory sample used here.

The fixed 15-topic integration experiment does not tune its topic count entirely within a nested supervised validation loop. A stronger future experiment would select that count inside each historical training fold and compare the integration variants on the same held-out dates. The current report keeps the integration claim limited to the observed test-window gain.